

INTERNET

MULTIMEDIA

SKM4263

L07 - Online Digital Storage

Dr. Norhidayah Azman



Lecture Outline



Overview of digital archiving

Examples of online digital storage

Threats to permanence

Security and privacy issues



Definition



a.k.a. Cloud Computing

**Online
Digital
Storage**

Internet/hosted storage:

Use a service provider to store data online,
Avoid storing data locally on physical disks

Used as:

Add-on method for existing data storage solutions,
Better access for increasing number of people



ODS Examples



ODS Examples

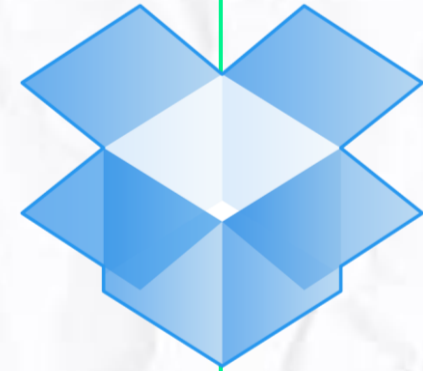


**Physical
location?**

**How it
works?**



Infra?



Advantages of ODS



Store personal documents

Store media related files

Advantages

Backup copies

People on the move



Disadvantages of ODS



**Internet
speeds**

**Limited storage
space**

Disadvantages

**Lack
of security**

**Service
charges**



Who is this?



SELLING

myMalaysia Citizen | 4M Data from myIDENTITY

by [redacted]

September 24, 2021 at 03:35 PM



New User

MEMBER

Posts

8

Threads

3

Joined

Apr 2020

Reputation

0

1 YEAR OF SERVICE

September 24, 2021 at 03:35 PM

Malaysia citizen data fresh from Jabatan Pendaftaran Negara (JPN), leaked from hasil.gov.my through myIDENTITY API.

Total data is almost 4 million equal to 31.8GB, group by birth year from 1998 to 1979.

Include

- name
- email
- mobile number
- permanent address
- gender
- NRIC
- picture in base64 String
- race
- religion

No of Files : 19
File Format: JSON / CSV
File size : 31.8GB GB

Price: 0.2 BTC

INSIDER

HOME > TECH

533 million Facebook users' phone numbers and personal data have been leaked online

Aaron Holmes 14 hours ago

- The personal data of over 500 million Facebook users has been posted online in a low-level hacking forum.
- The data includes phone numbers, full names, location, email address, and biographical information.
- Security researchers warn that the data could be used by hackers to impersonate people and commit fraud.

© Dr Norhidayah Azman 2026

Russia hacked MH17 investigation in Malaysia

By Naveen Goud

1009



The Dutch government has disclosed yesterday that the laptop used by expelled spies belonging to Russia intelligence contained evidence that the device was used to hack the investigation of Malaysian Airlines flight MH17 which is suspected to have crashed into the Indian Ocean in 2014.



WHY VC must resign:

1. Being a Political Puppet
2. Infringing Academic Freedom
3. Promoting racism & hatred
4. Failing to protect student's constitutional rights.
5. Failing to solve UM's financial crisis.

Rank	Twitch Streamer	Total Payout from August 2019 until October 2021		
1	CriticalRole	\$9,626,712.16		
2	xQcOW	\$8,454,427.17		
3	summit1g	\$5,847,541.17		
4	Tfue	\$5,295,582.44		
5	NICKMERCs	\$5,096,642.12		
6	ludwig	\$3,290,777.55		
7	TimTheTatman	\$3,290,133.32		
8	Altoar	\$3,053,839.94		
9	auronplay	\$3,053,341.54		
10	LIRIK	\$2,984,653.70		
11	__unknown__	\$2,863,780.63		
12	Gaules	\$2,844,985.18		
13	HasanAbi	\$2,810,480.11		
14	Asmongold	\$2,551,618.73		
15	loTyler1	\$2,490,584.90		
16	RanbooLive	\$2,401,021.84		
17	MontanaBlack88	\$2,391,369.58		
18	KnowNothingTV	\$0.00		
19	ibai	\$2,314,485.53		
20	Castro_1021	\$2,311,021.81		
21	MOONMOON	\$2,236,043.55		
22	TheRealKnessi	\$2,157,258.23		
23	moistcr1tikal	\$2,098,742.63		
24	Mizkif	\$2,086,548.21		
25	CohhCarnage	\$2,061,059.29		
26	shroud	\$2,040,503.15		

File: [bezos-pogchamp.jpg](#) (9 KB, 186x295)



☐ **twitch leaks part one** Anonymous 10/06/21(Wed)05:34:18 No.83691438 [▶](#) [>>83691646](#) [>>83691709](#) [>>83691771](#) [>>83691793](#) [>>83691840](#) [>>83691889](#) [>>83692116](#) [>>83692133](#) [>>83692140](#) [>>83692215](#) [>>83692493](#) [>>83692530](#) [>>83692648](#) [>>83692676](#) [>>83692742](#) [>>83692816](#) [>>83692928](#) [>>83693046](#) [>>83693122](#) [>>83693130](#) [>>83693202](#) [>>83693499](#) [>>83693675](#) [>>83694121](#) [>>83694305](#) [>>83694487](#) [>>83694534](#) [>>83694826](#) [>>83695179](#) [>>83695224](#) [>>83695340](#) [>>83695439](#) [>>83695568](#) [>>83695773](#)

We bring to you today an extremely poggers leak:

Twitch is an American video live streaming service that focuses on video game live streaming, including broadcasts of esports competitions, operated by Twitch Interactive, a subsidiary of Amazon.com, Inc.

Their community is also a disgusting toxic cesspool, so to foster more disruption and competition in the online video streaming space, we have completely pwned them, and in part one, are releasing the source code from almost 6,000 internal Git repositories, including:

- > Entirety of twitch.tv, with commit history going back to its early beginnings
- > Mobile, desktop and video game console Twitch clients
- > Various proprietary SDKs and internal AWS services used by Twitch
- > Every other property that Twitch owns including IGDB and CurseForge

- > An unreleased Steam competitor from Amazon Game Studios
- > Twitch SOC internal red teaming tools (lol)

AND: Creator payout reports from 2019 until now. Find out how much your favorite streamer is really making!

Torrent (128GB) [\[REDACTED\]](#)

Jeff Bezos paid \$970 million for this, we're giving it away FOR FREE.

#DoBetterTwitch

Threats to Permanence



Application Obsolescence



Corruption



Completeness



Findability



Threats to Permanence



Application Obsolescence

Old technology
replaced by
newer versions

Materials created
on outdated
technology
become no longer
accessible

File formats and
applications must
also be compatible
with replacement
systems

If digital records
are not converted
to new formats
→ risk of loss by
obsolescence

Example?



Threats to Permanence



Corruption

Digital materials made up of “bits”

- series of digits representing the material’s info/content
- read by a device / computer
- displayed or communicated for the human eye

If a bit is corrupted → entire record becomes corrupted

- One alteration in bits can result in the record’s effective death

Example?



Threats to Permanence



Completeness

Metadata →
structured info
that describes,
explains, and/or
locates an e-file

- E.g. file name, creation date, modification history
- Help make records become more comprehensible/relevant

Metadata should
be kept in a
manifest/
spreadsheet

- Future users know where a file is located and its purpose

Example?



Threats to Permanence



Findability

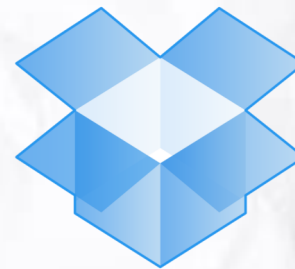
Digital files can become essentially lost and useless if

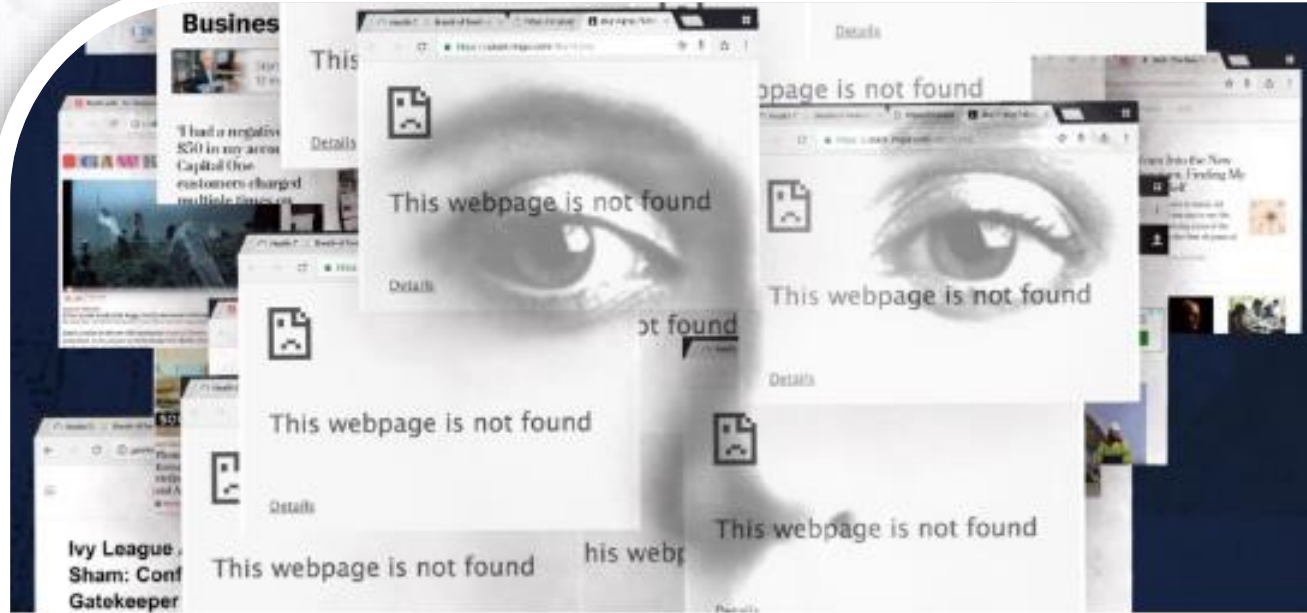
- one cannot locate them
- multiple copies are on different drives

Employ conscientious file-naming customs

- Consistent file-naming helps users organize and locate records

Example?





Columbia Journalism Review.

Erasing history

When an online news outlet goes out of business, its archives can disappear as well. The new battle over journalism's digital legacy.

Selecting an ODS



Make sure you thoroughly
check the contract text
before agreeing to it

Make sure you are the
exclusive owner of the
account

How to Choose?

Find out where the storage
facilities are located

Use unique usernames and
access passwords





Facebook moves 1.5bn users out of reach of new European privacy law

Company moves responsibility for users from Ireland to the US where privacy laws are less strict



General Data Protection Regulation (GDPR)

Summary



Digital archiving is made possible via ODS

Lots of ODS examples

Advantages and disadvantages of ODS

Threats to permanence of digital material

Lots of cybersecurity breaches and privacy infringements

Criteria to select ODS





Dr Norhidayah Azman



dayah@usim.edu.my



Dr Dayah



019-3513524